

DUEBENDORF AP, Switzerland

WMO: 068070

Lat: 47.402N

Lon: 8.634E

Elev: 435

StdP: 96.21

Time Zone: 1.00 (EUC)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-8.2	-5.8	-11.1	1.5	-7.6	-8.2	2.0	-4.1	9.1	9.1	7.5	7.2	1.7	120	0.461

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	31.2	20.3	29.2	19.6	27.2	18.9	21.3	29.0	20.2	27.1	19.6	26.0	2.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	14.4	24.6	17.9	13.6	23.6	17.1	12.8	22.4	63.8	29.4	60.4	27.6	57.7	26.0	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 7.4	6.1	5.0	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	-12.1	23.2	3.3	0.9	-14.5	23.9	-16.5	24.4	-18.3	24.9	-20.7	25.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.6	2.0	1.5	5.8	10.7	14.2	18.6	20.2	19.2	15.5	11.0	5.9	2.6
	DBStd	7.58	4.05	4.30	3.85	3.89	3.75	3.51	3.23	3.30	2.89	3.58	3.55	3.51
	HDD10.0	1062	248	238	136	39	7	1	0	0	1	33	130	229
	HDD18.3	3001	506	472	388	231	137	39	18	29	94	228	373	487
	CDD10.0	1298	1	0	6	58	137	258	316	286	165	64	7	1
	CDD18.3	196	0	0	0	0	8	46	77	57	8	1	0	0
	CDH23.3	1842	0	0	0	20	104	462	698	489	68	2	0	0
	CDH26.7	574	0	0	0	1	19	147	243	157	7	0	0	0
Wind	WSAvg	1.6	1.9	1.6	1.9	1.7	1.8	1.6	1.7	1.4	1.4	1.3	1.4	1.6
Precipitation	PrecAvg	1207	69	72	75	88	131	136	140	132	97	88	90	89
	PrecMax	1541	170	162	227	175	243	222	257	236	208	159	195	185
	PrecMin	950	9	21	24	12	30	51	53	18	45	8	0	1
	PrecStd	154	42	38	44	43	58	44	54	52	40	40	49	47
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.9	14.8	19.9	25.2	28.9	33.2	34.0	32.9	27.2	22.2	17.2	13.0
		MCWB	10.3	8.1	11.2	14.4	18.0	21.9	21.1	20.7	19.2	15.5	12.2	9.0
	2%	DB	11.9	12.0	17.1	23.1	26.0	30.2	31.8	30.2	25.2	20.2	15.1	11.1
		MCWB	8.4	7.1	9.9	13.5	16.9	19.8	20.2	20.4	18.1	15.1	11.5	8.5
	5%	DB	9.8	9.9	15.0	21.1	23.9	28.0	29.2	28.2	23.1	18.2	13.0	9.8
		MCWB	7.2	6.2	9.1	12.7	15.8	19.2	19.5	19.9	16.9	14.3	10.0	7.3
	10%	DB	7.2	7.2	12.8	18.9	21.8	25.9	27.2	26.0	21.1	16.2	11.1	7.9
		MCWB	5.5	4.7	8.1	11.7	14.5	18.3	18.7	19.2	16.2	13.3	9.0	6.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.4	9.0	11.7	15.3	18.6	23.1	22.2	22.2	19.9	16.7	13.0	10.0
		MCDB	13.4	14.3	18.4	23.5	26.8	31.6	30.7	29.2	25.7	20.9	17.0	11.7
	2%	WB	8.8	7.5	10.3	14.2	17.4	21.1	21.1	21.2	18.7	15.7	11.5	8.8
		MCDB	11.4	11.3	16.0	21.7	24.6	28.6	29.3	28.6	24.2	19.1	14.7	10.9
	5%	WB	7.2	6.2	9.4	13.0	16.3	19.7	20.1	20.2	17.6	14.7	10.4	7.5
		MCDB	9.2	9.2	14.1	20.4	22.0	26.5	27.0	26.7	22.3	17.7	13.0	9.5
	10%	WB	5.6	5.0	8.3	12.0	15.2	18.7	19.4	19.4	16.7	13.8	9.3	6.0
		MCDB	7.0	7.1	12.3	18.1	20.4	24.9	26.0	25.4	20.5	16.1	11.1	7.3
Mean Daily Temperature Range	5% DB	MDBR	5.7	7.1	9.5	10.8	10.3	10.7	10.5	10.1	9.5	8.2	6.3	5.8
		MCDBR	8.2	12.3	14.0	15.7	14.5	14.8	14.6	14.0	12.4	11.5	9.7	8.6
		MCWBR	5.7	7.9	8.1	7.8	6.7	6.2	5.4	5.9	6.3	6.8	6.6	6.0
	5% WB	MCDBR	7.9	11.2	12.7	14.4	12.8	13.4	13.0	12.6	11.2	10.6	9.3	8.1
		MCWBR	5.6	7.4	7.5	7.4	6.1	5.8	5.3	5.6	5.9	6.6	6.3	5.8
Clear-Sky Solar Irradiance	taub	0.292	0.317	0.353	0.389	0.403	0.408	0.411	0.403	0.376	0.349	0.317	0.288	
	taud	2.441	2.373	2.335	2.271	2.276	2.304	2.292	2.329	2.384	2.448	2.483	2.452	
	Ebn at Noon	799	846	864	863	861	857	849	840	830	799	761	758	
	Edh at Noon	66	87	106	125	129	126	126	116	101	81	64	59	
All-Sky Solar Radiation	RadAvg	1.18	2.03	3.24	4.34	4.98	5.72	5.54	4.83	3.63	2.29	1.28	0.97	
	RadStd	0.13	0.27	0.44	0.63	0.50	0.53	0.50	0.49	0.37	0.25	0.16	0.15	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (117 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		N/A	N/A	N/A	+0.71	+0.42	+0.34	N/A	N/A	+57	+16

Nomenclature: See separate page